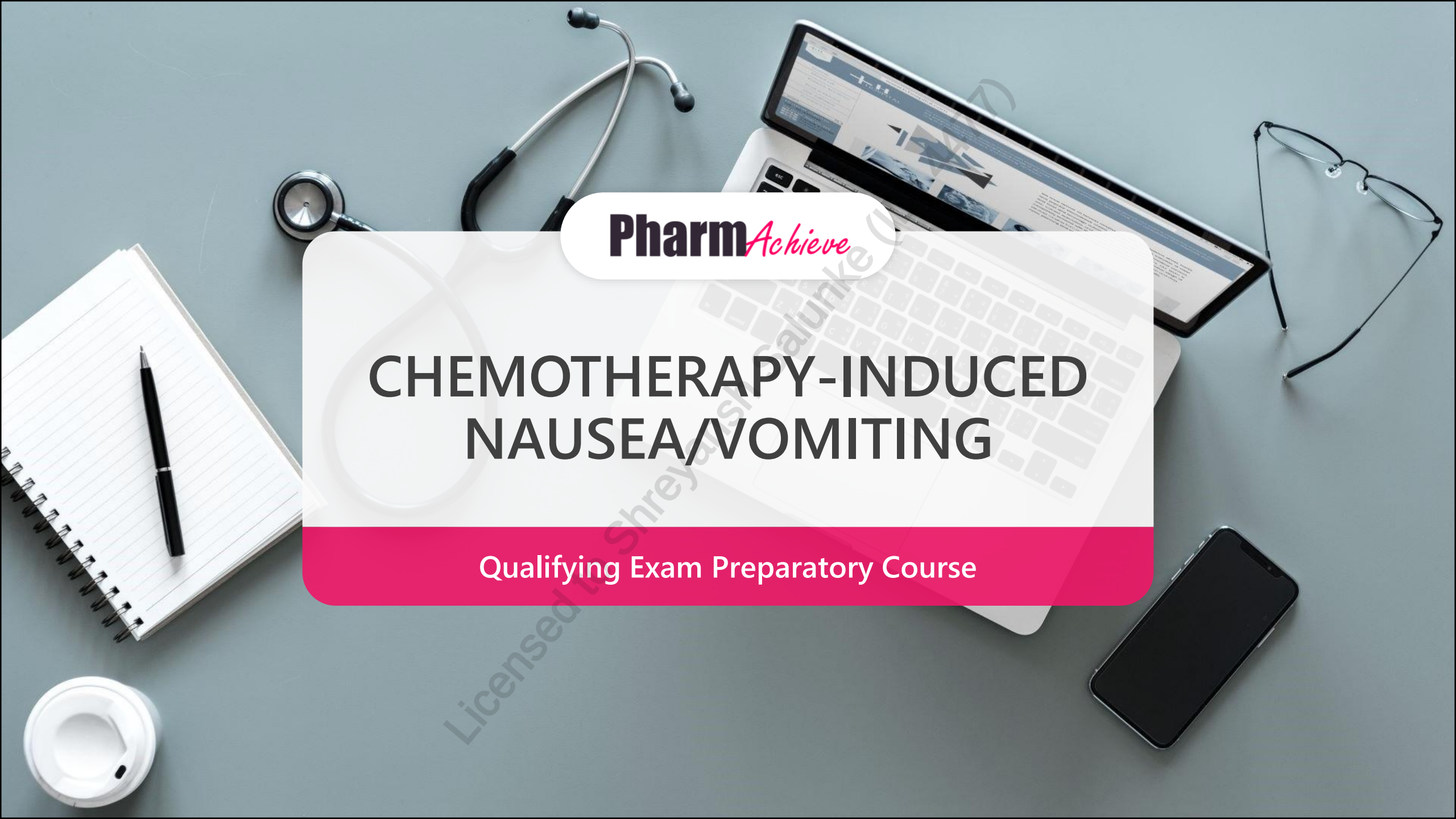


A top-down view of a grey desk with various items: a stethoscope, a laptop displaying a medical website, a tablet, a pair of glasses, a smartphone, a spiral notebook with a pen, and a white coffee cup. A large white pill-shaped graphic is centered over the laptop screen.

Pharm*Achieve*

CHEMOTHERAPY-INDUCED NAUSEA/VOMITING

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

**Providing Care:
Clinical Care**

**Providing Care:
Distribution**

**Knowledge
& Expertise**

**Communication
& Collaboration**

**Leadership &
Stewardship**

Professionalism

LEGEND

5-HT Serotonin

AC Combination of Doxorubicin (Adriamycin) and Cyclophosphamide

APR Aprepitant

AUC Area Under the Curve

BSA Body Surface Area

CINV Chemotherapy-Induced Nausea/Vomiting

CTZ Chemoreceptor Trigger Zone

DOP Dopamine Antagonist

DEX Dexamethasone

GI Gastrointestinal

LFTs Liver Function Tests

MCP Metoclopramide

MRI Magnetic Resonance Imaging

NCCN National Comprehensive Cancer Network

NEPA Netupitant and Palonosetron Combination Pill

N/V Nausea/Vomiting

NK1 Neurokinin 1 Receptor Antagonist

OLZ Olanzapine

SCC Squamous Cell Carcinoma

N/V DEFINITIONS AND CLINICAL PRESENTATION

N/V consists of 3 different mechanisms

Nausea: inclination to vomit, or feeling in the throat or epigastric region that vomiting is imminent

Retching: labored movement of abdominal and thoracic muscles without expulsion of stomach contents

Vomiting: expulsion of stomach contents through the mouth due to GI retro-peristalsis

N/V can occur transiently with no other associated signs and/or symptoms but can also be part of a more complex clinical presentation

PHASES OF CINV

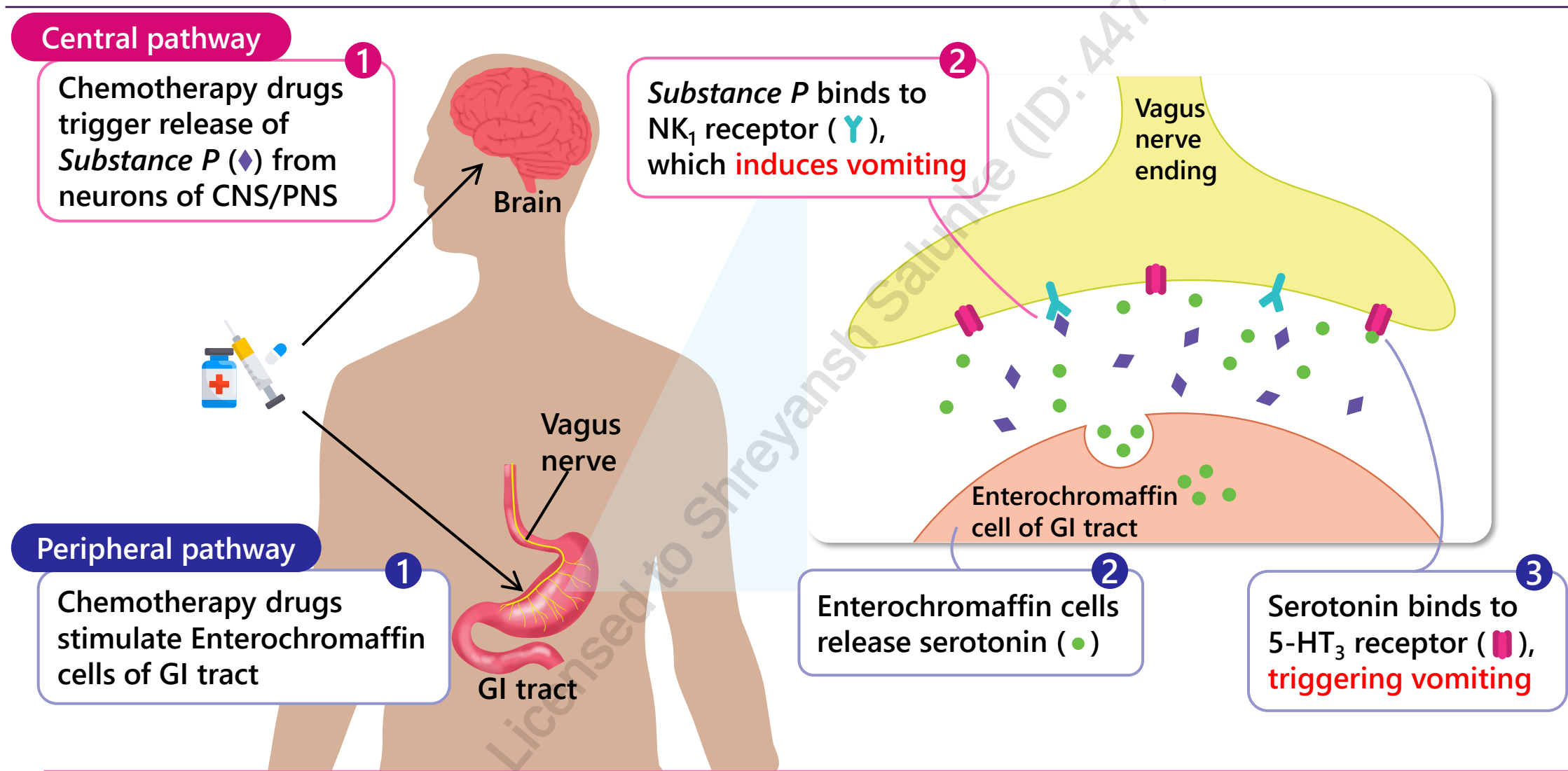
ACUTE PHASE: Occurs within 24 hours of chemotherapy administration

DELAYED PHASE: Begins > 24 hours after chemotherapy administration

BREAKTHROUGH: Occurs despite preventative treatment, may require rescue therapy

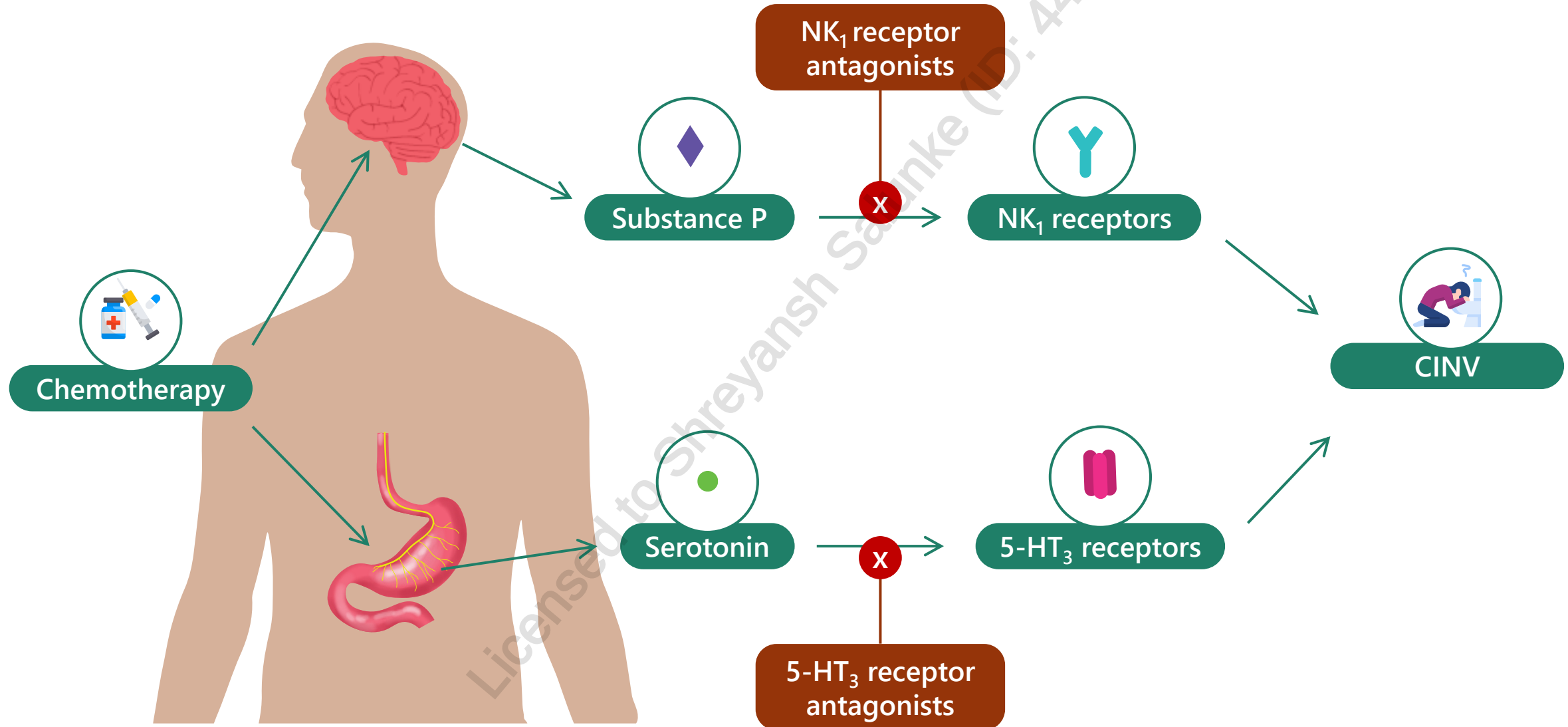
ANTICIPATORY EMESIS: psychological response occurring before chemo due to prior negative CINV experience

CHEMOTHERAPY-INDUCED NAUSEA AND VOMITING



LEGEND: ◆ = Substance P Y = NK_1 receptor ● = Serotonin ■ = $5-HT_3$ receptor

CHEMOTHERAPY-INDUCED NAUSEA AND VOMITING



PATHOPHYSIOLOGY

Patient receives chemotherapy

1. 5-HT released from stimulation of enteroendocrine cells from the GI tract
2. 5-HT interaction with 5-HT₃ receptors in intestinal wall
3. Afferent fibers stimulate CTZ
4. Stimulation of vomiting center
5. Stimulation of salivation/respiratory centers and GI/abdominal muscles

Patient experiences N/V

Substances involved:

- Dopamine
- Serotonin
- Substance P
- Histamine
- Acetylcholine
- Gamma-aminobutyric acid (GABA)
- Cannabinoids

RISK FACTORS

- Emetogenicity of chemotherapy
- Age
 - Younger age > older age
 - <50 years old = higher risk
- Sex
 - Women > Men
- History of Alcohol Use
 - Low alcohol intake > high alcohol intake
- History of N/V
 - Motion sickness and/or nausea during pregnancy
 - Poor CINV control with previous chemotherapy cycle

DIFFERENTIAL DIAGNOSIS

RULING OUT OTHER POSSIBLE CAUSES

Fluid and electrolyte abnormalities

- Hypercalcemia
- Hyponatremia
- Hypoglycemia or hyperglycemia
- Volume depletion
- Water intoxication
- Adrenocortical insufficiency

Drug-induced

- Opiates
- Antibiotics
- Antifungals

Other

- Constipation
- GI obstruction
- Dyspepsia
- Increased intracranial pressure
- Peritonitis
- Metastases: Brain, meninges, hepatic
- Uremia
- Infections (septicemia, local)
- Radiation therapy
- Pain

DIAGNOSIS

MEDICAL/DRUG RELATED CAUSES

Emetic Risk Group	Antineoplastic Agents (<i>note: not all drugs are listed</i>)
Minimal risk (<10%)	Bevacizumab, Bleomycin, Cladribine, Pembrolizumab, Rituximab, Trastuzumab, Vinblastine, Vincristine, Vinorelbine, Degarelix, Fulvestrant, Lanreotide, Octreotide, Goserelin
Low risk (10–30%)	Azacitidine, Docetaxel, Etoposide, 5-Fluorouracil, Gemcitabine, Mitomycin, Mitoxantrone, Paclitaxel, Pemetrexed, Topotecan
Moderate risk (>30–90%)	Bendamustine, Carboplatin (AUC<4), Cyclophosphamide ($\leq 1500 \text{ mg/m}^2$ of BSA), Doxorubicin (<60 mg/m^2), Idarubicin, Irinotecan, Oxaliplatin, Temozolomide
High risk (>90%)	Anthracycline/Cyclophosphamide combination, Carboplatin (AUC ≥ 4), Cisplatin, Cyclophosphamide (>1500 mg/m^2), Dacarbazine, Doxorubicin ($\geq 60 \text{ mg/m}^2$)

Adapted from NCCN Antiemetic Guidelines 2023

GOALS OF THERAPY

The goal is to prevent all 3 phases of CINV:

- Acute
- Delayed
- Anticipatory

Parameter	Degree Of Change	Timeframe
Post-chemotherapy N/V (CINV)	Eliminate or greatly reduce	Following each session

GENERAL PRINCIPLES OF MANAGING CINV

- Preventing CINV is preferable to treating (especially to prevent anticipatory CINV)
- The emetogenic potential of the chemotherapy regimen along with patient-specific risk factors will determine the preventative antiemetic regimen the patient receives
 - When chemotherapy regimens contain multiple drugs, the drug in the highest emetic risk group will determine which antiemetic regimen is chosen
- Patients should always have an antiemetic prescribed for breakthrough CINV
- Consider both the efficacy of the preventative (scheduled) and breakthrough antiemetics

CINV

Non-pharmacological options

- ✓ Eat small, light meals
- ✓ Avoid foods with strong aroma
- ✓ Eat bland foods
- ✓ Avoid triggers
- ✓ Stay well hydrated

Breakthrough and Anticipatory

In addition to scheduled antiemetics, patients should also be offered:

1. Breakthrough CINV treatment
2. An additional agent to be used if needed

Anticipatory CINV treatment

- Benzodiazepines, such as lorazepam, may be given before chemo
- Controlling acute and delayed CINV is the best approach to prevent anticipatory CINV

CINV PROPHYLAXIS

Emetic Risk	Prevention of ACUTE CINV		Prevention of DELAYED CINV
High	5-HT3 + DEX + NK1 + OLZ →		DEX + NK1* + OLZ
Moderate	One of:	5-HT3 + DEX →	5-HT3 or DEX
		5-HT3 + DEX + OLZ →	OLZ
		5-HT3 + DEX + NK1 →	NK1* ± DEX
Low	5-HT3 or DEX or DOP		None
Minimal	None		None

Important Note:

This table shows NCCN's "preferred" regimens. In practice, regimens may vary based on patient-specific factors and provincial coverage.

NK1* indicates that if aprepitant was used as the NK1 receptor antagonist for acute CINV on day 1, it should be given on days 2 and 3 as well. If netupitant (as part of NEPA) was given, no NK1 is needed for delayed CINV prevention.

TREATMENT OF BREAKTHROUGH CINV

Preferred Treatment	• Add a medication from a different drug class than what is currently part of the patient's antiemetic regimen • The preferred agent for breakthrough CINV is olanzapine (if not already used as part of the preventative regimen) • Antiemetics for breakthrough CINV can be added on a scheduled or as-needed basis
Additional Antiemetic Options	• Multiple drugs may be needed to treat breakthrough CINV • Options include prochlorperazine, promethazine, haloperidol, lorazepam, metoclopramide, ondansetron, granisetron, dexamethasone, nabilone, etc.

5-HT3 ANTAGONISTS

Example	• Ondansetron, Palonosetron, Granisetron (all can be given PO or IV)
MOA	• Antagonists of the 5-HT3 receptor • Block presynaptic serotonin receptors in gut wall • Partially block acute phase of CINV
Place in Therapy	• 1st line for preventing acute phase CINV and radiation-induced N/V • Not as good for delayed phase CINV (except palonosetron due to high affinity) • All 5-HT3 antagonists are shown to be effective, but some patients benefit from switching within the same class
Side Effects	• Headaches, dizziness, constipation, diarrhea, increased LFTs, QT prolongation

5-HT₃ ANTAGONISTS

PALONOSETRON PROPERTIES

- Renally and hepatically excreted
- **Prolongs QT interval**
- Administration: 1 dose 1 hour prior to chemotherapy (on day 1 of each cycle)
- Some evidence for being superior to other 5-HT₃ antagonists for patients receiving highly emetogenic chemotherapy
- Has 100-fold higher affinity to 5HT receptors compared to ondansetron and granisetron
- Longest half life ≈ 40 h (≈ 4 h ondansetron; $\approx 5-9$ h granisetron)
- **Causes significant constipation at dose of 0.75 mg**

CORTICOSTEROIDS

Example	Dexamethasone, Prednisone, Methylprednisolone
MOA	Unknown
Place In Therapy	1st line for CINV and radiation-induced N/V - Effective for both acute and delayed phases - Can be used alone for chemo with low emetic risk - Used in combo with other agents for high emetic risk chemo - Not used for simple N/V
Side Effects	- Acute: GI upset, hyperglycemia, fluid retention, mood changes, insomnia (take in daytime to minimize insomnia) - Chronic (>2-4 weeks): hypothalamic-pituitary-adrenal (HPA) axis suppression, Cushing's syndrome
Drug Interaction	When combined with aprepitant, reduce dexamethasone dose by 50% (CYP3A4 interaction)

SUBSTANCE P/NEUROKININ 1 (NK1) RECEPTOR ANTAGONISTS

Example	• Aprepitant (Emend® PO) , Fosaprepitant (Emend® IV), Netupitant
MOA	• Substance P binds NK receptor to cause N/V • Acute phase CINV mediated by serotonin and substance P • Delayed phase CINV mediated mostly by substance P • Aprepitant blocks NK receptor → as a result substance P cannot bind
Place in Therapy	• 1st line for high emetic risk chemotherapy • Usually combined with 5-HT3 antagonists + dexamethasone
Drug Interactions	• These agents are substrates and moderate inhibitors of CYP3A4 • Aprepitant and fosaprepitant also induce CYP2C9 • If given with dexamethasone, reduce dexamethasone dose by 50%

ATYPICAL ANTIPSYCHOTICS

Example	• Olanzapine (Zyprexa®)
MOA	• Blocks multiple receptors involved in CINV including dopamine, serotonin, and histamine
Place in Therapy	• 1st line for acute and delayed CINV for highly emetogenic chemotherapy regimens • Preferred agent for breakthrough CINV • Studied in doses 5-10 mg PO daily
Drug Interactions	• Enhance CNS depressant effects with other CNS depressants
Side Effects	• Extrapyramidal side effects, sedation, postural hypotension, constipation, hyperglycemia

PHENOTHIAZINES

Example	• Prochlorperazine, Chlorpromazine
MOA	• Block dopamine receptors in the CTZ
Place in Therapy	• Prophylaxis for patients on low-emetogenic chemo drugs • Used for breakthrough N/V • Many dosage forms are available, and all are equally effective • Practical for long-term treatment
Drug Interactions	• Enhance CNS depressant effects with other CNS depressants
Side Effects	• Extrapyramidal side effects, sedation, slurred speech, dry mouth, constipation, urinary retention, hyperprolactinemia

BUTYROPHENONES

Example	• Haloperidol
MOA	• Block dopaminergic stimulation of CTZ
Place in Therapy	• Effective, but usually only used in palliative setting • Droperidol is rarely used due to the risk of potentially fatal QT prolongation and Torsade de Pointes

BENZAMIDES

Example	• Metoclopramide
MOA	• Dopaminergic antagonist at standard doses • 5-HT antagonist at high doses • Increases lower esophageal sphincter tone • Aids gastric emptying • Accelerates transit through small intestine
Place in Therapy	• Can be used for prophylaxis of acute CINV with low emetic risk chemotherapy regimens • Can also be used as needed for breakthrough CINV
Side effects	• Extrapyramidal symptoms (dose-related), drowsiness, dizziness, diarrhea

ANTI-HISTAMINE-ANTICHOLINERGICS

Example	• Dimenhydrinate, Scopolamine*
MOA	• Interrupt visceral pathways that stimulate N/V
Place in Therapy	• Simple N/V
Side Effects	• Drowsiness, confusion, blurry vision, dry mouth, urinary retention, tachycardia
Contraindications	• Patients with narrow-angle glaucoma, prostatic hyperplasia, COPD

* Scopolamine currently only available as an injection on the Canadian market

CANNABINOIDS

Example	• Nabilone (oral)
MOA	• Cannabinoid receptor agonists, exact mechanism unknown
Place in Therapy	• Used as last resort if no other agents are effective • Used for breakthrough N/V
Side Effects	• Dizziness, confusion, "high" feeling, sedation, dry mouth, headache, mood alterations, hallucinations

BENZODIAZEPINES

Example	Lorazepam
MOA	Potentiate inhibitory action of GABA by binding to GABA_A receptors
Place in Therapy	Mostly used for anxiety, and therefore in anticipatory nausea
Side Effects	Sedation, anterograde amnesia, ataxia

MONITORING PARAMETERS

EFFICACY ENDPOINTS		
Parameter	Degree of Change	Timeframe
Anticipatory N/V	Eliminate or reduce	24-48 hours prior to chemotherapy
Acute phase N/V		Within 24 hours of chemotherapy
Delayed phase N/V		> 24 hours of chemotherapy
Breakthrough N/V		Throughout chemotherapy

SAFETY ENDPOINTS		
Parameter	Degree of Change	Timeframe
ADRs (e.g. drowsiness, headaches, hyperglycemia)	Prevent or minimize	Following each session
More serious ADRs (e.g. QT prolongation, extrapyramidal side effects)	Prevent	

TAKEAWAY

- ✓ Prevent CINV with appropriate antiemetics based on the emetic risk associated with the patient's chemotherapy regimen
- ✓ Anticipatory N/V is best prevented with proper control of acute and delayed CINV
- ✓ Antiemetics to prevent acute CINV are typically given on day 1 of chemotherapy, while those to prevent delayed CINV are given for the next few days post-chemotherapy
 - ✓ Exception: NEPA (netupitant/palonosetron) is only given prior to chemo on day 1, but still helps to prevent delayed CINV as both drugs have long half-lives
- ✓ Review patient's previous CINV history to determine how to optimize therapy for future CINV prevention and treatment
- ✓ Educate patient on the importance of taking antiemetic drugs pre-chemotherapy and how the medications work on CINV

CASE 1

NK is a 65-year-old post-menopausal female who presents to the hospital with a 2-month history of nasal obstruction symptoms, epistaxis, and tension headache. An MRI was completed, showing an extensive nasopharyngeal mass extending into the posterior nasal cavity, with multiple involved lymph nodes. A biopsy confirmed the diagnosis of SCC of the nasopharynx. She is receiving high-dose cisplatin with radiation for treatment. She has a past medical history of osteoarthritis. She currently takes acetaminophen 500 mg PO Q4-6H PRN (usually takes up to 4 tablets daily). She has no allergies to medications. She quit smoking 8 years ago (previously smoked 20 pack-years) and she drinks 1 alcoholic drink per week.

1. Which of the following is not a risk factor for CINV for NK?
 - a) Older age
 - b) Low alcohol intake
 - c) Highly emetogenic chemotherapy regimen
 - d) Female sex



CASE 1

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2. Antiemetics should be chosen with the goal of preventing which type of CINV?
- a) Acute
 - b) Delayed
 - c) Anticipatory
 - d) Acute, delayed and anticipatory CINV



CASE 1

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3. Which pre-chemotherapy anti-emetics should NK receive to prevent acute CINV?
- a) Ondansetron alone
 - b) Ondansetron + dexamethasone
 - c) Ondansetron + aprepitant + dexamethasone
 - d) Ondansetron + aprepitant + dexamethasone + olanzapine



REFERENCES

1. National Comprehensive Cancer Network. NCCN Clinical Practice Guidelines in Oncology: Antiemesis. 2023. <https://www.nccn.org/guidelines/guidelines-detail?category=3&id=1415>.
2. Basch E, Prestrud AA, Hesketh PJ, et al. Antiemetics: American Society of Clinical Oncology Clinical Practice Guideline Update. *JCO*. 2011;29(31):4189-4198. doi:10.1200/JCO.2010.34.4614
3. Chemotherapy-induced nausea and vomiting. In: eCPS. Ottawa, ON: Canadian Pharmacists Association. <https://myrx.tx.ca>
4. Hesketh P. Pathophysiology and prediction of chemotherapy-induced nausea and vomiting. UpToDate. <https://www.uptodate.com>.
5. Multinational Association of Supportive Care in Cancer. MASCC/ESMO Antiemetic Guideline 2023. <https://mascc.org/resources/mascc-guidelines/>
6. Management of side effects of chemotherapy and radiation therapy. In: RxTx. Ottawa, ON: Canadian Pharmacists Association.
7. ASCO 2017 Guidelines. <http://ascopubs.org/doi/abs/10.1200/JCO.2017.74.4789>
8. Janelins MC, Tejani M, Kamen C, et.al. Current Pharmacotherapy for Chemotherapy-Induced Nausea and Vomiting in Cancer Patients. 2013;14(6):757-766. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3938333/#!po=45.8333>
9. Adel, N. Overview of Chemotherapy-Induced Nausea and Vomiting and Evidence-Based Therapies. 2017. <https://www.ajmc.com/journals/supplement/2017/managed-care-considerations-in-chemotherapy-induced-nausea-and-vomiting/overview-of-chemotherapy-induced-nausea-and-vomiting-and-evidence-based-therapies-article?p=2>
10. Cancer Care Ontario. 2019 Antiemetic Recommendations for Chemotherapy-Induced Nausea and Vomiting: A Clinical Practice Guideline. <https://www.cancercareontario.ca/sites/ccocancercare/files/guidelines/full/2019AntiemeticRecommendationsChemotherapyInducedNauseaVomiting.pdf>
11. National Comprehensive Cancer Network. NCCN Guidelines Version 2.2024 Antiemesis. 2024. https://www.nccn.org/professionals/physician_gls/pdf/antiemesis.pdf

CHANGE LOG

August 2025

- Slide 11: Removed streptozocin since no longer on Canadian market
- Slide 22: Removed brand names since no longer on Canadian Market
- Slide 23: Removed droperidol since no longer on Canadian market
- Slide 25: Added note on scopolamine availability as an injection only
- Slide 31: Changed option D to better align with PEBC exam question style

September 2025

- Slide 28: Elaborated on monitoring parameters

March 2026

- Slide 2: Updated NAPRA competencies
- Slide 11: Minor update in table title
- Formatting changes made throughout